

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

LECTURE 12.1

Power Grid Stability Estimation

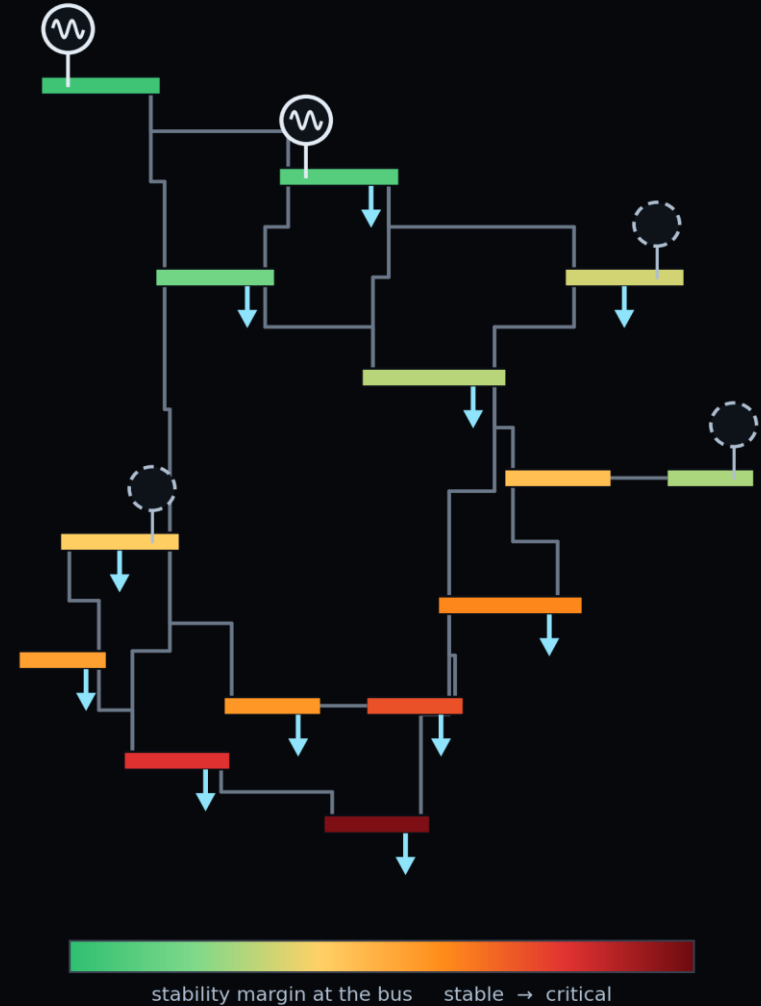
The state problem: what is the grid doing right now, when you can only measure part of it. Algebraic first, then dynamic — and the physics is what fills the gaps.

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AC power flow · weighted least squares · the swing equation · PMU data



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **write** — the power flow equations, and say why they are constraints and not a simulation
- **explain** — observability as a property of the graph, not of sensor quality
- **formulate** — estimation as a residual the network is judged against, not a map it l
- **build in** — the slack angle and what is certain; check the voltage band
- **identify** — inertia and damping from a recorded frequency disturbance
- **say** — where the estimate fails, and what that failure looks like

WHAT TODAY ASSUMES

- | | |
|------------|--|
| from L5.1 | graph networks, and message passing |
| from L7.1 | residual, composite loss, hard enforcement |
| from L8.2 | identifying a parameter from a curve |
| from L11.2 | ODEs in a loss, not PDEs |

TWO HALVES, TWO OPERATORS

Algebraic constraints for the steady state, then the swing equation — same machinery.

AND REAL DATA

Energinet and Fingrid publish enough to do both halves from measurement.

From a Car You Can Hold to a Machine You Cannot

WHAT CHANGES, AND WHAT DOES NOT

- L11 ended with a model checked against an instrument you could lift onto a bench
- **the grid inverts that** — the equations are the same kind of object, the access is not
- so the question stops being 'is my model right' and becomes 'what can I know from four measurement points'
- which is exactly the shape a physics residual is good at

THE SAME MACHINERY

- | | |
|------------------------------|---------------------------------|
| L7 hard constraints | the slack bus, voltage limits |
| L8 inverse problems | identifying inertia and damping |
| L9 residual weighting | measurements against physics |
| L11 identifiability | reappears, and bites harder |

WHAT IS GENUINELY NEW

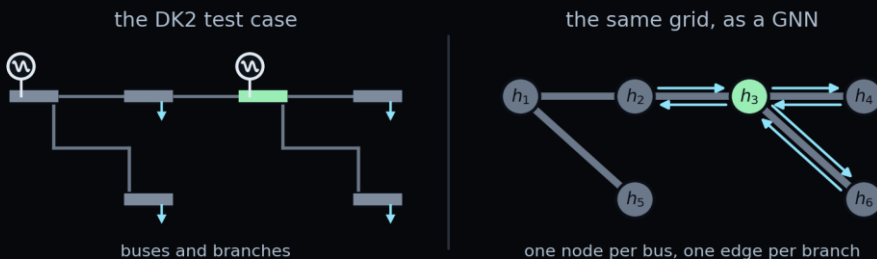
The system is a network, so the unknowns are coupled through a graph rather than through space. A missing measurement at one bus is repaired by its neighbours, and how well depends on the topology, not on distance.

That is the idea worth taking away, and it is the one the exercise is built around.

A Physics-Informed Graph Network

WHAT YOU ALREADY HAVE

- L5.1 covered graph networks alongside CNNs, and this is where they are appropriate here
- **a grid is not a grid of pixels** — the neighbours are whoever the edges connect
- so message passing, not convolution, and the adjacency comes from the topology



THE NETWORK, CONCRETELY

inputs	per-bus features, plus the adjacency
hidden	3 to 5 message-passing layers
output	voltage magnitude and angle per bus
from L5.1	permutation invariance, locality

WHY NOT AN MLP ON A FLAT VECTOR

Its input dimension is tied to one topology. A graph network takes the topology as an argument.

ALGEBRAIC NOW, DIFFERENTIAL LATER

A grid needs both: algebraic power flow, and ODEs at the machines.

A Grid Is a Graph With Ohm's Law on Every Edge

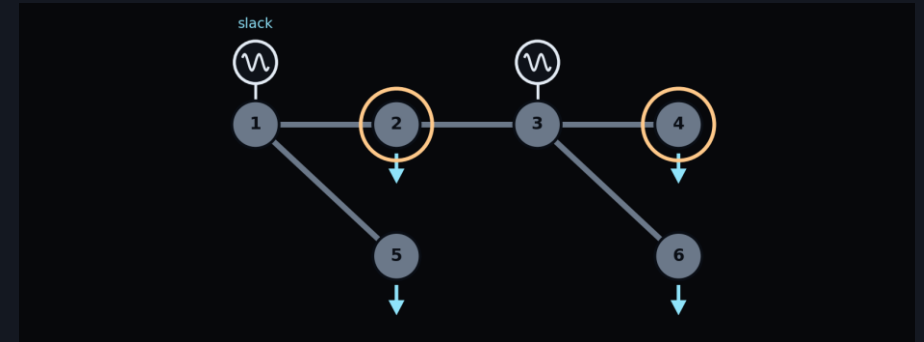
BUSES, BRANCHES, AND ONE MATRIX

- a bus is a node; a branch is an edge with an impedance
- the admittance matrix is built from topology and line parameters, both of which the operator knows
- nothing about it is learned
- **the unknowns are the voltages** — a magnitude and an angle at every bus

$$S_i = P_i + jQ_i = V_i \sum_k V_k Y_{ik}^* e^{j\theta_{ik}}$$

S_i the power injected at bus i , V the voltages, Y_{ik} the admittances, θ_{ik} the angle differences

SIX BUSES SHOW EVERYTHING



WHY ANGLES, NOT PHASES

Only angle differences appear in any power equation, so one bus must be the reference, at angle zero. Skip it and the problem has a one-parameter family of solutions that any optimiser will wander along.

The Power Flow Equations

REAL AND REACTIVE INJECTION AT EVERY BUS

$$P_i = V_i \sum_k V_k (G_{ik} \cos \theta_{ik} + B_{ik} \sin \theta_{ik})$$

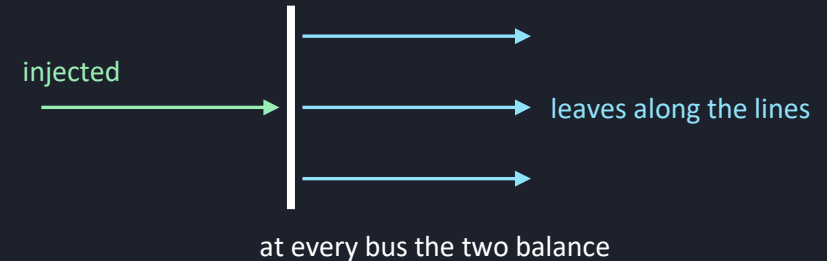
$$Q_i = V_i \sum_k V_k (G_{ik} \sin \theta_{ik} - B_{ik} \cos \theta_{ik})$$

P_i, Q_i active and reactive power at bus i ; $G_{ik} + jB_{ik}$ the admittance Y_{ik}

- two equations per bus, nonlinear in the voltages and trigonometric in the angle differences
- these are constraints, not a simulation
- they hold at every instant, in steady state, whatever the disturbance history
- this is the physical content of the algebraic half of the lecture

THE STATE

$$x = \{V_i, \theta_i\}_{i=1}^n, \quad \theta_{ik} = \theta_i - \theta_k$$



THE REFERENCE, FIXED BY HAND

$$\theta_1 = 0, \quad V_1 = 1 \text{ p.u.}$$

Enforce this as you enforced a Dirichlet condition in L7: build it into the trial solution rather than penalising it. A slack bus only approximately at zero is not doing its job.

What a Control Room Actually Sees

TWO MEASUREMENT SYSTEMS, DECADES APART

- **SCADA has been there since the 1960s** — injections and flows, scanned every few seconds
- phasor measurement units are the modern layer, reporting angle directly, tens of times a second
- PMUs are expensive, so they are sparse
- a real network has them at a minority of buses and a full SCADA layer underneath

SCADA

quantities	P, Q injections and flows; some $ V $
rate	one scan every 2–4 s
timing	unsynchronised across substations
angle	not measured at all

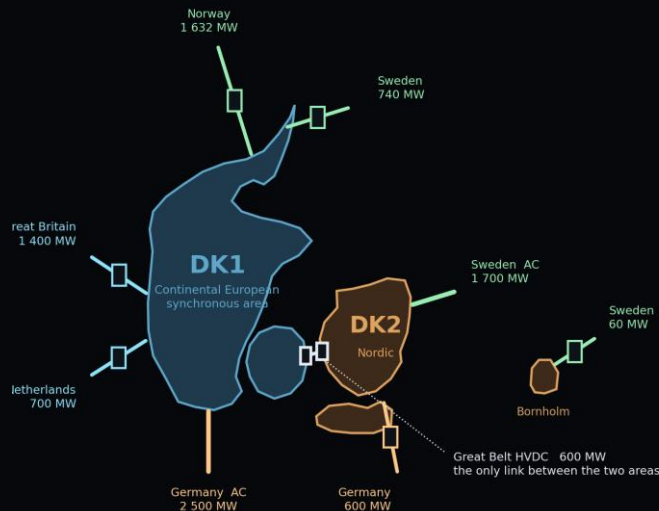
PMU

quantities	V and I phasors, magnitude and angle
rate	10–120 samples per second
timing	GPS-synchronised, microsecond class

Two Grids in One Small Country

DENMARK IS SPLIT, AND THAT IS USEFUL

- western Denmark is in the Continental European synchronous area; eastern Denmark is Nordic
- so two halves of one country run at different frequencies



WHAT THAT MEANS FOR THE MODEL

DK2	Nordic synchronous area, AC to Sweden
DK1	Continental area, AC to Germany
Great Belt	HVDC — power, not synchronism
Kontek	HVDC, DK2 to Germany
consequence	an HVDC link is a boundary condition

HVDC IS A BOUNDARY, NOT A BRANCH

A DC link does not appear in the swing coupling at all. It enters as a scheduled injection you impose, which is convenient: it lets you study one synchronous area without modelling its neighbours, provided you say that is what you did.

What Is Public, and What Is Not

REAL OPERATING DATA IS FREELY AVAILABLE

- Energinet publishes production, consumption and interconnector flows per price area
- Fingrid and Svenska kraftnät publish the Nordic area frequency, historically and live
- between them you have both halves of this lecture from measurement rather than simulation
- which is unusual, and worth exploiting

THE ONE THING YOU CANNOT DOWNLOAD

No TSO publishes a nodal model with line impedances — topology and parameters are withheld for security. So the network you compute on is representative, not actual. Say that once, clearly, and stop apologising for it.

SOURCES WORTH KNOWING

Energi Data Service	DK1/DK2 production and load
Fingrid open data	Nordic frequency, live
Svenska kraftnät	the same, independently
ENTSO-E Transparency	pan-European load
PyPSA-Eur	an open European topology
IEEE test cases	small, exact, for verification

Nothing here is Denmark-specific as a method. Swap in any TSO's open data and any network model — the exercise is written so the grid is an input, not an assumption.

Fewer Measurements Than Unknowns

THE MEASUREMENT MODEL

$$z = h(x) + e, \quad \dim z = m, \quad \dim x = 2n - 1$$

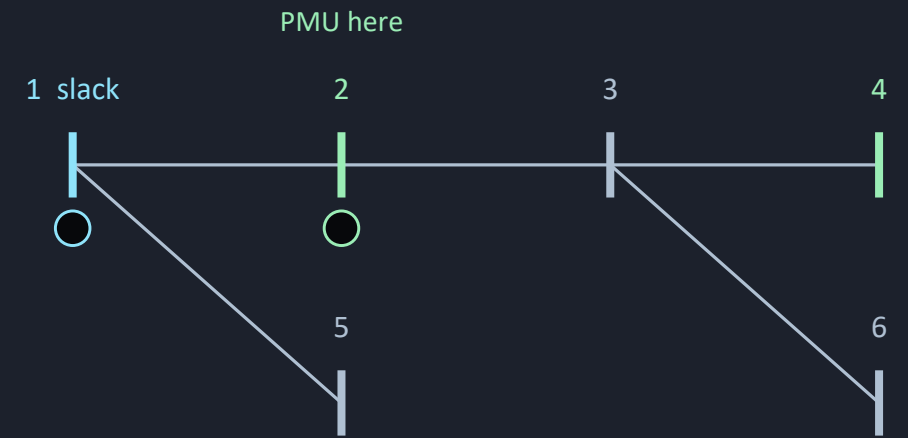
z the measurements, h the power flow expressions, e the error, n buses

- the measurement model is the power flow expressions plus error
- when measurement points outnumber unknowns the system is redundant and errors average down
- when they do not, it is under-determined and no amount of fitting repairs that
- the classical answer is pseudo-measurements** — forecasts entered as if they were measurement points

THE HONEST FRAMING

A state estimator does not measure the grid. It produces the state most consistent with the readings you have and the physics you believe, and the second half of that sentence is doing at least as much work as the first.

READ AT TWO BUSES, WANTED AT ALL SIX



Weighted Least Squares, and What It Costs

THE ESTIMATOR EVERY UTILITY ALREADY RUNS

$$\hat{x} = \operatorname{argmin}_x \sum_{j=1}^m \frac{(z_j - h_j(x))^2}{\sigma_j^2}$$

z_j measurement j , $h_j(x)$ the model's prediction of it, σ_j its standard deviation, m their number

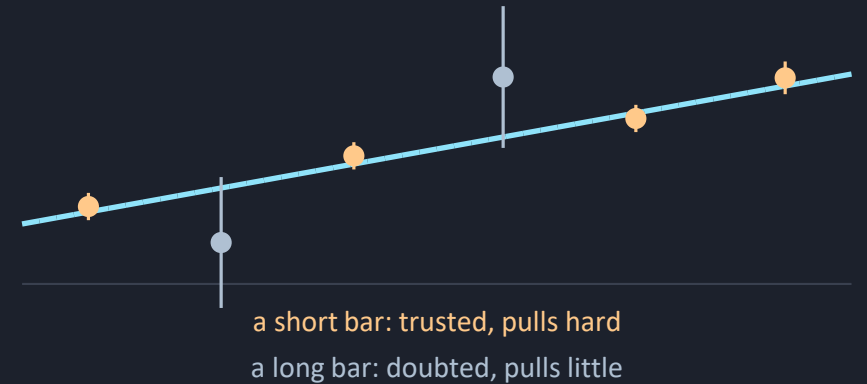
- weight each residual by how much you trust that measurement point, and minimise
- solved by Gauss–Newton, fast, understood, in production for fifty years
- any new method has to beat that, not merely work
- what it needs is a well-conditioned Jacobian, and thin measurements do not give one

$$(H^T R H) \Delta x = H^T R (z - h(x))$$

WHERE IT STRUGGLES

- thin instrumentation** ill-conditioned without pseudo
- bad data** one gross error smears across
- topology error** a wrong breaker voids the model
- fast dynamics** assumes a steady state

A FIT WEIGHTED BY TRUST



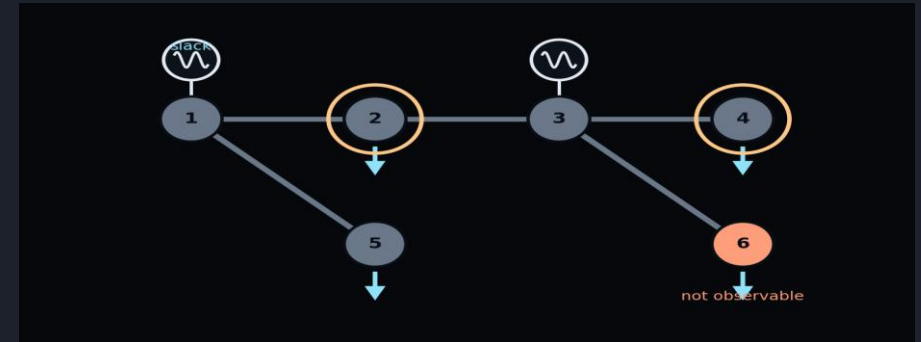
Observability Is a Property of the Graph

A RANK CONDITION, NOT A QUANTITY OF DATA

$\text{rank } H = 2n - 1 \Rightarrow \text{observable}$ H the Jacobian of the measurements
 n the number of buses

- observability asks whether the measurements determine the state at all
- it depends on where the measurement points are, not on how good they are
- classically the network splits into observable islands
- **a physics residual changes what is determinable** — the equations constrain the unmeasured buses too

UNOBSERVABLE, AND WHAT FILLS THE GAP



SAY WHICH IT IS

An estimate at an uninstrumented bus is an inference from the model, not a reading. The exercise asks you to report the two separately, because a plot that does not distinguish them invites the reader to trust both equally.

The Network Learns the State, Not the Physics

WHAT THE NEURAL NETWORK IS FOR

- the tempting formulation hands the network measurements and asks for the state
- the physics-informed one is different, and once seen the one used here
- the network proposes a state; the power flow equations judge it
- this is the same move as every exercise since L7

$$\mathcal{L} = \mathcal{L}_{\text{meas}} + \lambda_{\text{pf}} \mathcal{L}_{\text{pf}}$$

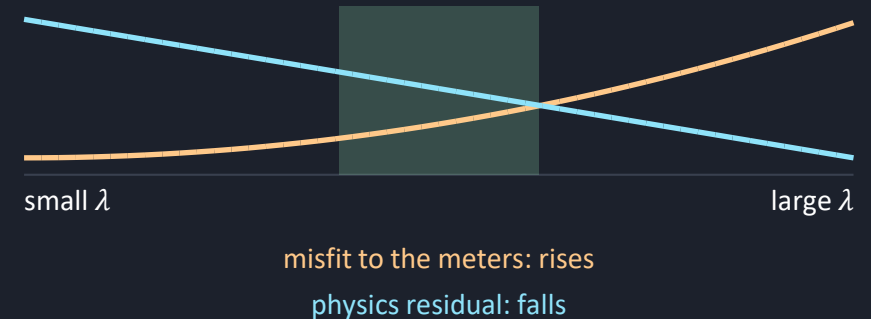
λ_{pf} the weight of the physics term
 $\Delta P_i, \Delta Q_i$ the power mismatch at bus i

$$\mathcal{L}_{\text{pf}} = \frac{1}{n} \sum_i (\Delta P_i^2 + \Delta Q_i^2)$$

TWO TERMS, ONE TRADE

data term	agree with the measurements from a meter, or from any solver
physics term	obey Kirchhoff everywhere
λ, the loss weight	model against measurements
no label term	no true state is needed

THE WEIGHT IS THE WHOLE ARGUMENT



The Architecture the Graph Asks For

MESSAGE PASSING, ON THE NETWORK YOU ALREADY HAVE

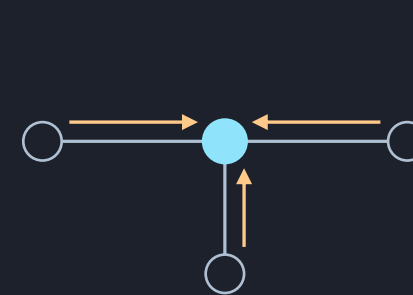
$$h_i^{(l+1)} = \sigma \left(W_1 h_i^{(l)} + \sum_{j \in \mathcal{N}(i)} W_2 h_j^{(l)} \right)$$

$h_i^{(l)}$ the features of bus i at layer l
 $\mathcal{N}(i)$ its neighbours, σ the activation
 W_1, W_2 weights shared by all buses

- each bus carries a feature vector, updated from itself and its neighbours
- three properties make it the natural fit
- permutation invariance, so renumbering the buses changes nothing
- locality, matching how the physics actually couples

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$$

WHAT MAPS ONTO WHAT



a node: a bus

an edge: a line

messages: from
the neighbours

node features: the readings at that bus; edge features: the line's admittance

THE MASK MATTERS AS MUCH AS THE DATA

Most buses have no measurement. Feed zeros and the network learns that zero is a reading; feed a mask channel alongside and it learns that the value is absent. The distinction between "measured as zero" and "not measured" is the whole problem in one design choice.

What a Graph Network Adds, and What It Does Not

A CLAIM WORTH EXAMINING BEFORE YOU MAKE IT

- the usual argument is that GNNs suit grids because they embed the topology
- true, and insufficient** — today's estimator fits one grid with one topology
- the real case appears when you change the formulation
- train across topologies, and the model amortises rather than re-solving

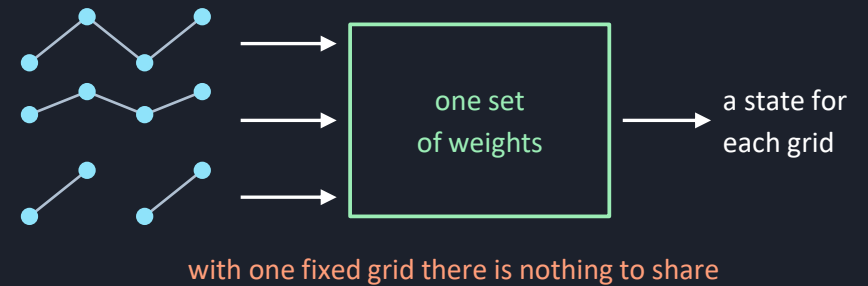
$$\hat{x} = g_{\theta}(z, \mathcal{G}), \quad \mathcal{G} = (\mathcal{V}, \mathcal{E})$$

g_{θ} the trained network, z its inputs at the buses, $\mathcal{G}$ the graph of buses $\mathcal{V}$ and branches $\mathcal{E}$

WHERE THE GNN GENUINELY WINS

changing topology	a line out, a bus split, N-1 cases
one model, many grids	weights shared, not refitted
speed at inference	a forward pass, no solve
large networks	cost grows with edges

ONE MODEL, MANY GRIDS



Build In What You Know for Certain

WHAT IS A LAW, AND WHAT IS AN OPERATING RULE

$$V_i = V_{\min} + (V_{\max} - V_{\min})\sigma(\hat{V}_i)$$

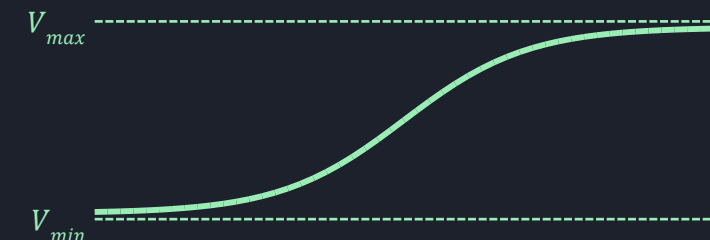
$\hat{V}_i$ the raw output, σ the logistic function
 $V_{\min}, V_{\max}$ the voltage band

- **the slack angle is a definition, not a measurement** — subtract it, never penalise it
- a scheduled voltage at a PV bus is given by the problem, so build it in the same way
- an operating band of 0.94 to 1.06 per unit is a rule, and a grid in trouble breaks it
- impose a rule and the estimate cannot show you the violation you needed to see

HARD, SOFT, AND NEITHER

slack angle	hard — a definition, exactly zero
PV bus setpoint	hard — given by the problem
power flow	soft — the residual, weighted by λ
voltage band	neither — check, never impose
line, Q limits	neither — check, never impose

BUILT IN, OR ONLY CHECKED



a value built in: it cannot leave its range
 an operating limit is checked, not built in

How You Know the Estimate Is Any Good

THREE CHECKS, IN ORDER OF STRICTNESS

$$\varepsilon = \frac{\|\hat{x} - x_{\text{true}}\|}{\|x_{\text{true}}\|}$$

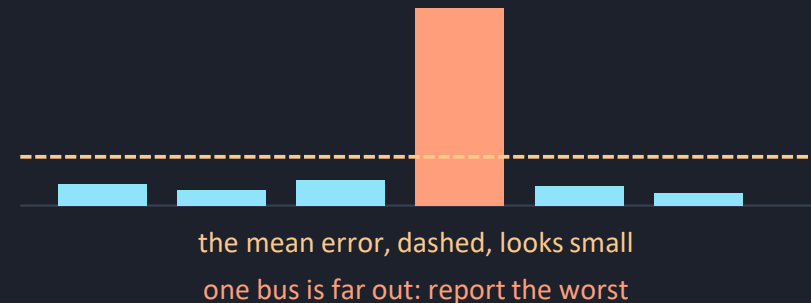
$\hat{x}$ the estimate, x_{true} the reference state, in the two norm

- on a synthetic case you know the truth, because you generated the measurements
- on real data you have none, so fall back on the residual
- and the third check is the one is often omitted
- **remove a measurement point and rerun** — a good estimator degrades gracefully

REPORT THESE SEPARATELY

instrumented	error where you had a reading
uninstrumented	error where physics alone carried it
worst bus	the mean hides the failure
one PMU dropped	error after removing one

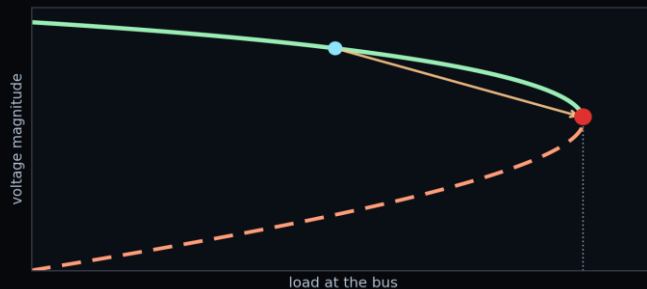
THE MEAN IS THE WRONG NUMBER



Now Let the Grid Move

EVERYTHING SO FAR ASSUMED A STEADY STATE

- the power flow equations describe a grid in balance
- they are exactly true when nothing is changing, and that assumption breaks first
- a line trips, a plant disconnects, and for the next few seconds nothing is in balance
- so the second half of this lecture lets the grid move



WHAT CHANGES WHEN t APPEARS

state	voltages, plus rotor angles and speeds
equations	algebraic constraints plus ODEs
data	PMU streams, not periodic scans
unknowns	inertia and damping too
failure	a machine losing synchronism

The machinery does not change. This is L8.2 after L8.1 — the same problem with a time derivative restored, and the same lesson about initial conditions waiting at the end of it.

The Swing Equation

A ROTATING MASS WITH AN ELECTRICAL BRAKE

$$\frac{2H_i}{\omega_s} \ddot{\delta}_i + D_i \dot{\delta}_i = P_{m,i} - P_{e,i}$$

δ_i the rotor angle, ω_s the synchronous speed
 H_i the inertia constant, D_i the damping
 P_m, P_e mechanical and electrical power

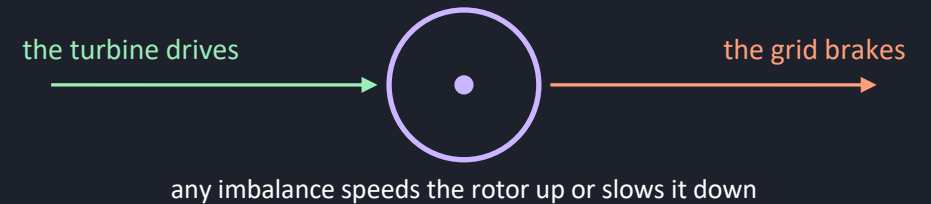
A synchronous generator is a heavy spinning rotor. Mechanical power comes in from the turbine, electrical power leaves through the network, and any imbalance accelerates or decelerates the rotor. That is the entire content of the equation.

H is the inertia constant — stored kinetic energy divided by rating, in seconds, typically two to eight. D lumps damping. Both are nominally known from the machine data sheet, and both drift from it in practice, which is what makes them worth estimating.

It is Newton's second law for a rotor, and it should feel like the longitudinal dynamics of L11.2 with the car replaced by a turbine. The identifiability problem is the same too: you cannot separate everything the data cannot distinguish.

AS A FIRST-ORDER SYSTEM

$$\dot{\delta}_i = \omega_i - \omega_s, \quad \dot{\omega}_i = \frac{\omega_s}{2H_i} (P_{m,i} - P_{e,i} - D_i \dot{\delta}_i)$$



WHAT IS BEING IDEALISED

no control response neither AVR nor governor acts

constant E excitation held fixed

classical model no sub-transient detail

network algebraic, infinitely fast

The Term That Ties the Machines Together

ELECTRICAL POWER OUT OF MACHINE i

$$P_{e,i} = \sum_k E_i E_k (G_{ik} \cos \delta_{ik} + B_{ik} \sin \delta_{ik})$$

E_i the internal voltage of machine i
 δ_{ik} the angle between i and k

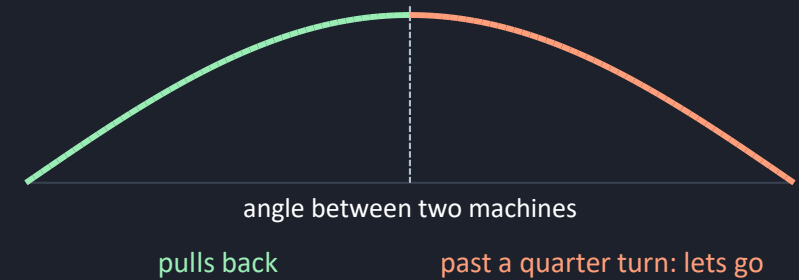
The electrical power depends on the angle differences between this machine and every other one. That single fact is why grid stability is a system property and not a machine property: no generator swings alone.

Note the shape of the coupling. It is sinusoidal in the angle difference, so it is restoring for small separations and stops being restoring once the difference grows past a quarter turn. A machine pushed beyond that does not come back.

Loads are folded into the network and eliminated, leaving an equivalent admittance between machine internal nodes. It is a strong simplification and the exercise says so, but it keeps the system small enough to see.

CENTRE OF INERTIA

$$\delta_{\text{COI}} = \frac{\sum_i H_i \delta_i}{\sum_i H_i}$$



FREQUENCY, AND ITS RATE

$$f_i = f_0 + \frac{\dot{\delta}_i}{2\pi}, \quad \text{RoCoF} = \frac{df_i}{dt}$$

RoCoF is what protection relays watch. It is the derivative of a noisy measurement — exactly what a physics-constrained estimate gives more cleanly.

What a Real Disturbance Looks Like

THE SHAPE YOU ARE FITTING

A large unit trips. Frequency falls at a rate set by the kinetic energy stored in every synchronous machine in the area — the initial RoCoF. Within seconds the fall arrests as reserves respond.

Three regimes, three physics, and only the first is the swing equation alone. Fit a window spanning all three with a model containing no control action and the mismatch lands in your inertia estimate.

So choose the window deliberately: the second or two after the event, where the machines coast and the governors have barely moved.



$$E_k = \sum_i H_i S_{n,i}, \quad \text{RoCoF}_0 = \frac{f_0 \Delta P}{2E_k}$$

$S_{n,i}$ the rating, ΔP the power lost
 f_0 the nominal frequency

READ THE EVENT IN THIS ORDER

0 to ~1 s	inertial — swing equation alone
1 to ~30 s	primary reserve arrests the fall
30 s onward	secondary control restores frequency
nadir	the lowest point, for protection
steady error	droop, not inertia

FALLING INERTIA, IN ONE EQUATION

Initial RoCoF is the power imbalance divided by twice the stored kinetic energy. Replace synchronous plant with converter-connected generation and the denominator falls: the same trip gives a steeper fall and less time to act.

Dynamic State Estimation, Same Three Moves

THE NETWORK IS NOW A FUNCTION OF TIME

The state becomes a trajectory: rotor angles and speeds against time, from a network taking t as input. Autograd supplies the time derivatives.

The measurement term compares against a PMU stream, not a scan. The dynamic residual asks the trajectory to satisfy the swing equations at collocation points in time.

The algebraic residual does not go away: a full formulation carries both, and the weights are again yours to justify.

$$\mathcal{L}_{\text{dyn}} = \frac{1}{N} \sum_{k=1}^N \|\dot{\mathbf{x}}(t_k) - \mathbf{f}(\mathbf{x}(t_k); \lambda)\|^2$$

$\mathbf{x}$ the machine state, $\mathbf{f}$ the swing equation in first order form, λ the parameters H_i, D_i

$$\mathcal{L} = \mathcal{L}_{\text{pmu}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{alg}} \mathcal{L}_{\text{pf}}$$

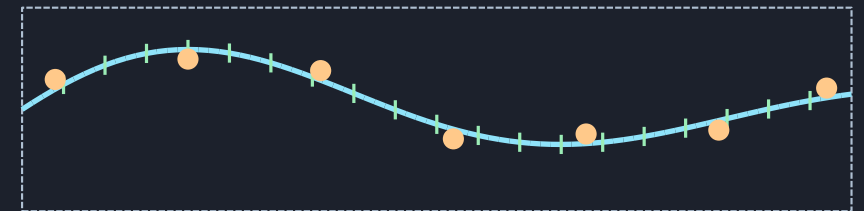
$\mathcal{L}_{\text{pmu}}$ the measurements, $\mathcal{L}_{\text{dyn}}$ the swing residual, $\mathcal{L}_{\text{pf}}$ the power flow residual

WHY NOT A KALMAN FILTER

Extended and unscented Kalman filters are the incumbent here and they are good. They are recursive, they carry a covariance, and they run in real time.

The physics-informed formulation is not recursive: it fits a window at once. A disadvantage in real time, an advantage for parameters, which the whole window constrains at once.

A WINDOW, FITTED AT ONCE



PMU readings, and the instants where the swing equation holds
one window, fitted at once

Inertia and Damping, Recovered From a Disturbance

PARAMETERS AS TRAINABLE VARIABLES

$\lambda = \{H_i, D_i\}$ trainable

H_i the inertia constant
 D_i the damping of machine i

Declare H and D trainable and minimise the same loss. This is the inverse-diffusivity exercise of L8.2 and the traction-constant identification of L11.2, with different letters — and by now it should look routine.

It matters because inertia is no longer a number you can look up. As synchronous plant is replaced by converters, a region's effective inertia falls, varies through the day, and is not instrumented. Operators would like to know it and do not.

The disturbance is what makes it identifiable. A system in steady state contains no information about H: nothing is accelerating, so the term multiplying H is zero. You need the grid to be disturbed, which is an uncomfortable requirement.

WHAT THE DATA CANNOT SEPARATE

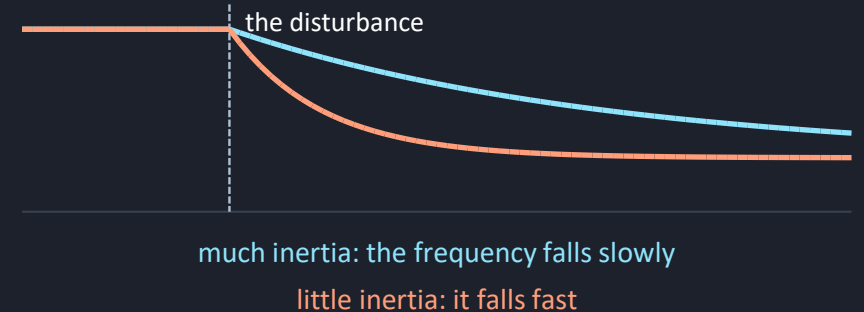
H and D only across timescales

H and P_m wrong P_m biases inertia

machines in a group coherent machines act as one

small disturbance H barely enters — distrust it

WHAT INERTIA DOES



What the Residual Contributes

THREE THINGS, AND ONLY THREE

First, estimates where there are no measurement points. The residual couples uninstrumented buses to instrumented ones through the network, so the state propagates outward. This is the headline claim and the one the exercise tests directly.

Second, noise rejection with a shape. Filtering assumes noise is fast and signal is slow; a physics residual instead assumes the trajectory satisfies a known equation, which is a much stronger and more specific prior. Derived quantities like RoCoF benefit most.

Third, parameters. A purely data-driven estimator produces a state; this produces a state and the physical constants that generated it, which is the thing an engineer can act on.

That is the list. It does not include speed, it does not include guarantees, and it does not include working when the model is wrong.

WHERE IT WINS

sparse instrumentation	the gap the method was built for
derived quantities	RoCoF, angle differences
parameter drift	inertia that is not on any sheet
offline analysis	a window, fitted at leisure

WHERE IT DOES NOT

real time	a filter is recursive, this is not
wrong topology	confidently wrong, everywhere
rich instrumentation	WLS is already fine, and faster
guarantees	no covariance falls out of it

How This Fails, and What It Looks Like

FOUR FAILURES YOU WILL MEET IN THE EXERCISE

THE WEIGHT IS WRONG

λ too high and the estimate is a power flow solution that ignores your measurement points; too low and you fit noise and the uninstrumented buses drift. Both look like convergence. The loss curve will not tell you — only the held-out measurement point will.

THE TOPOLOGY IS WRONG

A breaker status that does not match reality puts a wrong Y in the residual. The estimator then produces a beautifully consistent state of a network that does not exist, with small residuals throughout. This is the dangerous one.

THE DISTURBANCE IS TOO SMALL

Inertia enters through acceleration. A quiet window contains almost no information about H , and the optimiser will still return a value — one determined by your initialisation rather than by the grid.

THE MODEL IS TOO SIMPLE

The classical machine model omits excitation and governor response. Fit it to data from a real machine over a long enough window and the mismatch will be absorbed into H and D , which will come out physically implausible.

The Network You Will Compute On

REPRESENTATIVE TOPOLOGY, REAL OPERATING POINT

The exercise ships a small network whose structure follows DK2 in the ways that matter — a handful of load centres, generation concentrated at a few buses, and two HVDC links entering as fixed injections. Line parameters are plausible values, not measured ones.

Onto that structure you map real Energinet injections for a chosen hour, and real Nordic frequency for a chosen event. The topology is invented; the operating point and the dynamics are not.

This is a compromise and you should be able to defend it. The alternative — an IEEE test case with synthetic loading — is fully reproducible and entirely artificial. The alternative in the other direction is not available to anyone outside a control room.

The code takes the network as an input. Point it at an IEEE case, at a PyPSA-Eur extract, or at your own system, and everything downstream runs unchanged.

WHAT IS REAL AND WHAT IS NOT

injections	real — Energinet, per price area
frequency	real — Nordic area measurement
topology	representative, not published
line impedances	plausible, not measured
HVDC flows	real, imposed as boundary

THE HABIT THIS TEACHES

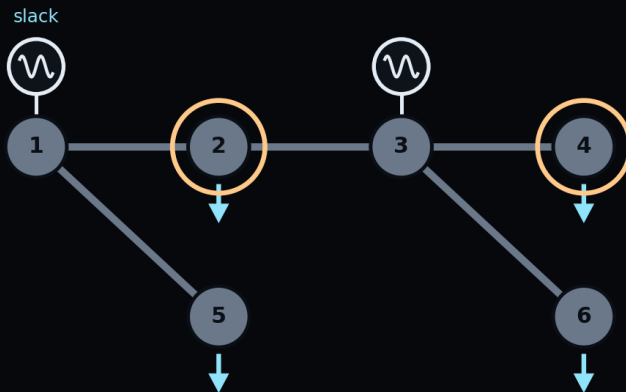
Every applied model mixes measured and assumed quantities. The discipline is not avoiding assumptions — it is labelling them, so a reader can see which conclusions would survive if one were wrong. L10.2 asked the same of you.

Ex_12.1 — What You Will Build

A SIX-BUS SYSTEM, THEN A DISTURBANCE

You start from a solved power flow on a small test system, generate noisy measurements from it, and hide most of them. The truth is available throughout, which is why you can measure what worked.

One notebook. Section 4 is the least-squares baseline: keep its numbers, because the network of section 5 is reported against them.



THE NOTEBOOK, BY SECTION

- | | |
|-------|---------------------------------------|
| 1 – 3 | the grid, its meters, the power flow |
| 4 | least squares on three sets of meters |
| 5 | the network, no true state — TODOs |
| 6 | inertia from a recorded disturbance |
| 7 | what the notebook says |
| 8 – 9 | your report, the mini projects |

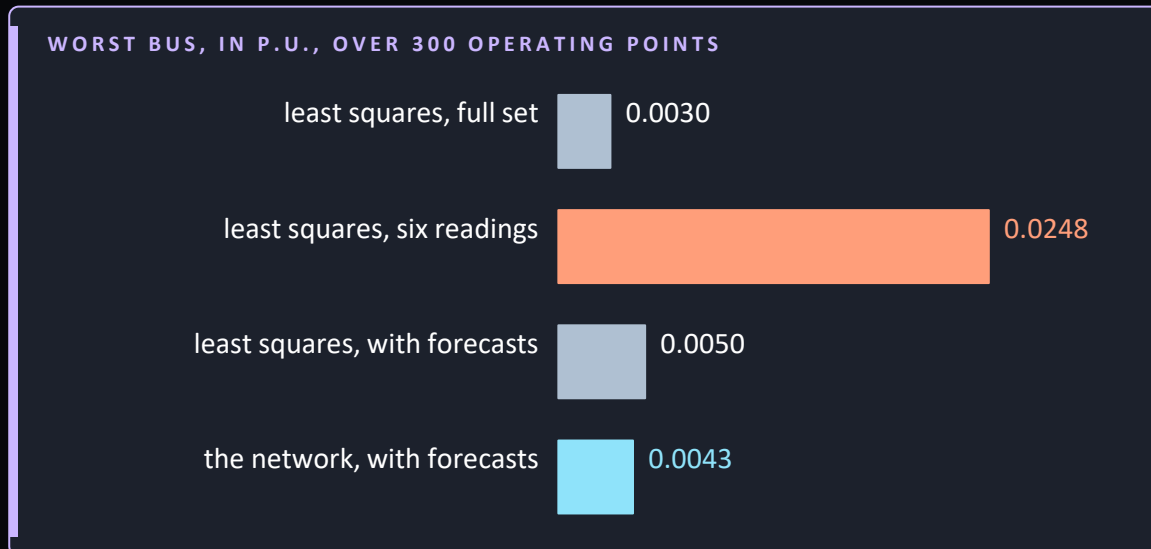
THE QUESTION THAT CARRIES THE MARKS

The report asks which of your numbers you would show an operator, which you would not, and why.

Roughly What the Numbers Look Like

ORDER OF MAGNITUDE, NOT A TARGET

From the same readings the two estimators agree closely. Least squares needs no training; the network, once trained, answers faster. That they agree is the control that says your implementation is correct.



WHAT MUST HOLD

full set	least squares within 0.003 p.u.
six readings	5 of 11 unknowns; off by 0.025
with forecasts	both within 0.005 p.u.
a line out, untold	both wrong; the check rises
quiet window	inertia untrustworthy — say so

If any of these five comes out backwards, something is wrong with the implementation and not with the physics. That is the debugging checklist.

Questions

1 – 2 · STATE ESTIMATION

1 What are the power flow equations, and what is state estimation?

Two algebraic balances at every bus, of active and of reactive power, written with the admittance matrix. They are constraints the voltages must satisfy, not a simulation in time. State estimation finds the voltage magnitude and angle at every bus from fewer, noisy measurements.

Ex_12.1 sections 3 – 4

2 What does observability depend on, and what does a physics-informed graph network add?

On where the measurements sit in the graph, not on how good the sensors are. The network outputs the state and is judged by the power flow residual at every bus, so a bus with no meter is still constrained. Weighted least squares stays the baseline to beat. *Ex_12.1 sections 4 – 5; L5.1*

3 – 4 · SWING DYNAMICS

3 What does the swing equation add to power flow?

Time. The rotor angle of each machine accelerates with the difference between its mechanical and its electrical power, held back by inertia and damping, and the electrical power ties the machines together through the grid. It is an ordinary differential equation in the loss. *Ex_12.1 section 6*

4 How are inertia and damping identified, and how is an estimate checked with no ground truth?

They are trained beside the weights so that the swing equation fits a recorded frequency disturbance, as the diffusivity was in L8.2. With no true state to compare against, the check is the residual map: where the physics is not satisfied, the estimate is not to be trusted. *Ex_12.1 sections 5 – 6; L8.2*

Where Should the Next Measurement Point Be Placed?

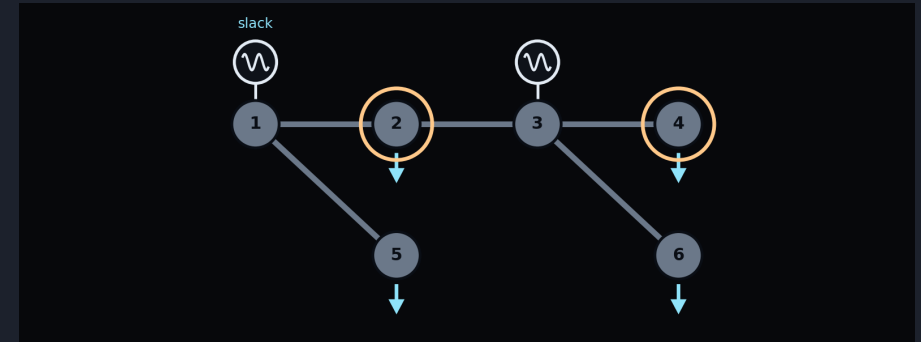
THE QUESTION THE METHOD LETS YOU ASK

Once you can estimate with sparse instrumentation, the natural question inverts: given a budget for one more PMU, where does it do the most good? That is a design question, and it is answerable by experiment — remove and add measurement points, and watch the worst-bus error.

The answer is rarely the busiest bus. It is usually the one that breaks an unobservable island, or the one furthest from existing coverage in the graph sense, and it depends on topology in ways that are hard to guess and easy to test.

This is an optional strand beyond the notebook, a loop of ten lines over its least-squares estimator, and the most open-ended thing here. It is also the closest the course comes to letting the model inform a decision rather than describe a state.

TRY IT BY EXPERIMENT



AN HONEST OPTIMISATION

Six buses means you can simply try every placement and rank them. Say that you did it exhaustively — a complete brute-force answer is worth more than a clever one that might have missed the optimum.

One Line to Remember

A physics residual turns "no measurement there" into "a weaker measurement there" — and the task is knowing how much weaker.

AND THREE THINGS THAT FOLLOW FROM IT

An estimate at an uninstrumented bus is an inference. Label it as one, every time, in every plot you hand to somebody who did not run the code.

The weight between measurement and physics is not a hyperparameter to tune away. It is the statement of how much you trust your model relative to your instruments, and it belongs in the report as a number with a justification.

A model you cannot check is not a measurement. The wrong-topology failure is the whole argument in one example.

WHAT L12.2 DOES WITH THIS

This lecture recovers the state now. The next one asks whether the system is heading somewhere it should not — the same equations integrated forward, with the disturbance you care about absent from the training data.

That last clause is the whole difficulty, and it is where a physics residual earns most clearly.

References and Further Reading

THE CLASSICAL SIDE

Schweppe & Wildes (1970), Power system static-state estimation, IEEE Trans. PAS — the paper that started the field.

Abur & Expósito (2004), Power System State Estimation — the standard reference for WLS, observability and bad-data detection.

Kundur (1994), Power System Stability and Control — the swing equation, machine models and transient stability.

THE PHYSICS-INFORMED SIDE

Raissi, Perdikaris & Karniadakis (2019), J. Comput. Phys. 378, 686–707 — the formulation used throughout this course.

Liu (2025), PINN with Python — the implementation conventions the exercises follow.

The power-system PINN literature is young; treat any accuracy claim in it as provisional until you have reproduced it.

READ IN THIS ORDER

if you are new to grids Kundur ch. 11, then Abur ch. 1–2

if you know grids Raissi et al., then the exercise

for the exercise the run guide in the core module

for L12.2 Kundur ch. 13, transient stability

Nothing in this lecture requires a paper you cannot get through the university library, and nothing in the exercise requires anything beyond torch, numpy and matplotlib.

NEXT

L12.2 — Prediction of Power Grid Stability

Estimation asks what the grid is doing. Prediction asks where it is going — and whether it comes back. Same equations, integrated forward, with the disturbance you care about missing from every training set you have.

BEFORE THE NEXT SESSION

run	Ex_12.1 sections 1 to 4
bring	your least-squares numbers
think about	where you would put one more PMU
read	Kundur ch. 13 if you can

The baseline matters. Everything in L12.2 is reported against something, and if you arrive without a number to compare to, you will spend the session generating one.